

## Introduction

### 1 in 7 people report using AI chatbots instead of contacting a GP 1 in 5 decided against professional advice on that basis

Sport & Exercise Medicine (SEM) is unusually exposed because its interventions span weeks/months without direct clinician supervision. This review investigated the use of AI in SEM and adjacent fields to:

- 1) **Appraise** the evidence base and regulatory landscape;
- 2) **Evaluate** opportunities and risks;
- 3) **Deliver** priorities for responsible development, validation & implementation.

## Results

139 sources from across the domains of:

- **Biomedicine:** SEM, Orthopaedics, Allied MSK
- **Informatics:** Digital health, software engineering, AI
- **Regulatory:** Legal & governance standards

Four domains of realistic application to SEM:

- **Treatment:** Injury management & rehabilitation
- **Prevention:** Injury & illness prevention
- **Decision support:** Optimising clinical management
- **Health data science** research, learning, 'meta-audit'

Classification of AI tools according to end user:

- **Provider:** clinics, services, clinicians, organisations
- **Service Users:** patients, athletes (supervised / unsupervised)
- **Non-clinical:** researchers, developers, administrators

**Poor evidence:** all contexts, especially in athletes, sport, nutrition  
**AI Chatbots:** widespread use (1 in 7 public), problematic in ~50%  
**Acceptability:** evidence of engagement decline if unsupervised  
**Governance:** mostly voluntary, little unified regulatory oversight  
**Consequence:** higher in unsupervised or decision-support tools  
**Priorities:** validation, stakeholder-led design, reporting standards

## Methods

### Search strategy

A structured literature search and analysis was conducted using *Aletheia* and *Oghma* – search, evaluation, and analytic engines built on a bespoke research platform under development by the group (Symposium, Aurora Performance, UK). This included:

- Conventional biomedical science databases
- Focused search of major organisations
- Citation searching & snowballing

### Included literature

Peer-reviewed articles, grey literature, frameworks & standards, legal & regulatory documents, real-world cases.

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## Applications, evidence and governance

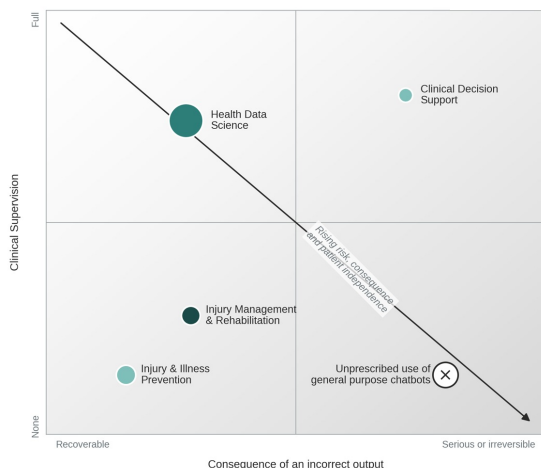
Artificial intelligence in Sport, Exercise & Musculoskeletal Medicine (SEM): what the evidence shows across four domains of use



|                             | TRIPOD+AI                                                                                       | PROBAST                                                      | TRIPOD-LLM                                                                    | CHART                                                      | SPIRIT-AI and CONSORT-AI                                                         | DECIDE-AI                                                          | NASSS                                                                         | FUTURE-AI                                                         | BS 30440:2023                                         | MHRA framework                                                  | Regulation (EU) 2024/1689                                                   | Lifecycle stage            |
|-----------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------|
| <b>Origin</b>               | Collins 2024                                                                                    | Wolff 2019                                                   | Gallifant 2025                                                                | Huo 2025                                                   | Cruz Rivera & Liu 2020                                                           | Vasey 2022                                                         | Greenhalgh 2017                                                               | Lekadir 2025                                                      | British Standards 2023                                | MHRA 2025                                                       | EU 2024                                                                     | 1 Purpose declaration      |
| <b>Purpose</b>              | Reporting of regression or machine-learning prediction model studies, development to validation | Risk of bias and applicability of prediction model studies   | Reporting of studies developing or evaluating large language models in health | Reporting of studies that evaluate chatbot health advice   | Trial protocols and trial reports for AI interventions                           | Early live clinical evaluation of AI decision support              | Explaining non-adoption, abandonment and failure to scale, spread and sustain | Consensus best practice across the lifecycle under six principles | Validation framework for AI products in healthcare    | How software and AI as a medical device are regulated in the UK | Statutory classification, human oversight and post-market monitoring of AI  | 2 Development              |
| <b>Type</b>                 | Reporting guideline                                                                             | Risk-of-bias tool                                            | Reporting guideline                                                           | Reporting guideline                                        | Reporting guideline                                                              | Reporting guideline                                                | Framework                                                                     | Framework                                                         | Certifiable standard                                  | Regulatory guidance                                             | Regulation                                                                  | 3 Internal validation      |
| <b>Strength</b>             | Complete for development, internal and external validation; widely endorsed                     | Four domains, twenty signalling questions; pairs with TRIPOD | Modular by task; covers evaluation of general-purpose models                  | Fixes reporting of prompt, model version and rating method | Requires algorithm version, input handling and human-AI interaction to be stated | The only instrument for first live use, human factors and safety   | Seven domains; treats the organisation and system, not the algorithm          | Reaches deployment and monitoring for drift                       | The only instrument a third party can certify against | Clarifies qualification, classification and post-market duties  | Enforceable duties and penalties, on deployers as well as providers         | 4 External validation      |
| <b>Limitation</b>           | Reporting, not appraisal; silent on a model in use                                              | Written for regression; judgement based; ends at validation  | New; uptake not yet measurable; reporting only                                | Excludes trials and prospective cohorts                    | Trial stage only; few AI trials exist in SEM                                     | Decision support only; not for tools acting on the person directly | Descriptive; no thresholds, no verdict                                        | Self-assessment; principles rather than tests                     | Certification cost; uptake unknown                    | Most consumer and wellness tools fall outside it                | High-risk duties deferred; general-purpose chatbots largely outside Annex I | 5 Early live evaluation    |
| <b>Applicability to SEM</b> | The benchmark injury-risk and prognosis models should meet                                      | Appraises the injury-prediction model literature             | Governs any study of chatbot advice on injury, exercise or nutrition          | Standard for auditing chatbots answering SEM questions     | The yardstick for the handful of rehabilitation trials                           | Covers first deployment of fracture-detection or triage tools      | Explains why engagement with patient-facing tools decays after adoption       | A lifecycle checklist for a group building its own tool           | A procurement criterion for tools entering a service  | Determines whether an injury app is a device at all             | Applies to any device-class tool marketed in Ireland or the EU              | 6 Randomised trial         |
|                             |                                                                                                 |                                                              |                                                                               |                                                            |                                                                                  |                                                                    |                                                                               |                                                                   |                                                       |                                                                 |                                                                             | 7 Deployment               |
|                             |                                                                                                 |                                                              |                                                                               |                                                            |                                                                                  |                                                                    |                                                                               |                                                                   |                                                       |                                                                 |                                                                             | 8 Post-market surveillance |

Legend:   
■ Voluntary  
■ Voluntary, certifiable by external audit  
■ Binding

Stage spans taken from each instrument's own scope statement; ten of eleven are voluntary



## Discussion

There is evidence of **potential benefits** of AI in SEM, e.g. clinical decision support; supported self-management; individualised care; health data science; data collection; service improvement & delivery.

We classify AI tools according to the end user: **Provider** (e.g. clinician); **Service User** (e.g. patient); **Non-clinical** (e.g. researcher).

**Regulation:** is almost entirely voluntary, and oversight is fragmented.  
**Risk** is highest when AI tools: requires unsupervised patient operation; are relied on for clinical decision-making, and when the consequence of error is serious or irreversible.

**Implementation** is most defensible where scale is the benefit and error is recoverable.

**Recommended priorities** for responsible design & implementation:

- Declared purpose** prospectively, which determines regulatory class;
- Validation of AI tool** performance in target population;
- Safety demonstration** with deliberate, specific, standardised testing;
- Professional body involvement** in guidance for use and regulation;
- Stakeholder involvement** in tool development, research, governance.

1. Bullock GS, et al. Sports Med 2022;52:2469-82.
2. Sumner J, et al. Artif Intell Med 2023.
3. Farmer L, et al. Health Technol Assess 2026.
4. Archer L, et al. Health Technol Assess 2026.